

# Teaching a Small LLM Scholastic Voice

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Independent Research — [github.com/pleyva2004/scholastic-llm](https://github.com/pleyva2004/scholastic-llm)

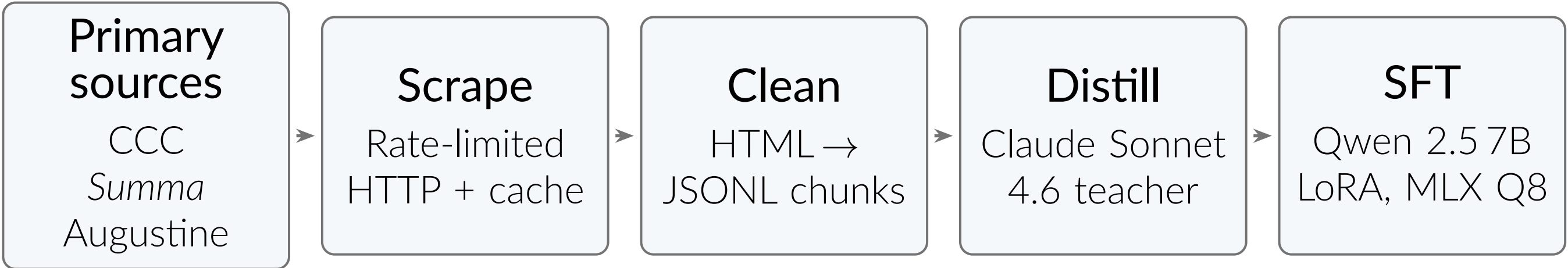
## Motivation

Modern instruction-tuned LLMs default to flat, encyclopedic prose. Specialized historical writing has a distinct lexical and structural fingerprint: the Summa’s *quaestio* form, Augustine’s autobiographical address, the Catechism’s numbered paragraph citations.

**Question.** Can a small open-weights model (7B parameters) be moved into this scholastic register *cheaply*, on a single laptop, using only public-domain or fair-use source texts?

**Answer.** Yes. With teacher-distilled supervised fine-tuning on ~80 (question, scholastic-answer) pairs, a rubric measuring register, citation grounding, and argument structure rises from **19/120 to 68/120** (+258%) on 10 held-out philosophical prompts.

## Pipeline



- Scrape CCC (vatican.va), Summa, Augustine (newadvent.org) with rate-limiting and on-disk cache.
- Clean into 342 structured JSONL chunks.
- Claude Sonnet 4.6 as *teacher*: per chunk, generate 2–3 (question, scholastic-answer) pairs with explicit register and CCC-citation instructions. 93 pairs, ≈\$0.50.
- Supervised fine-tune Qwen 2.5 7B-Instruct under MLX.

## Training Setup

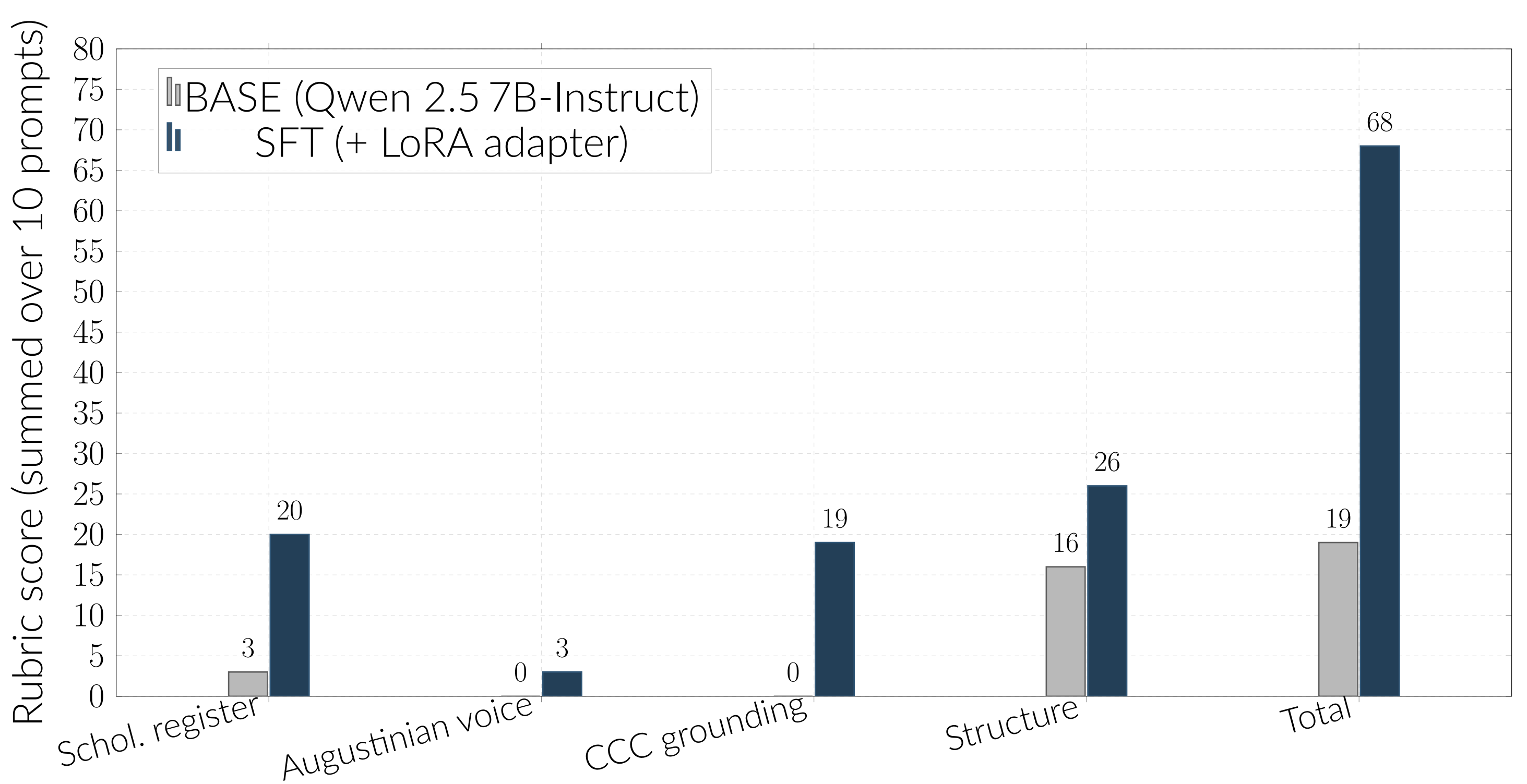
- Base: **Qwen2.5-7B-Instruct**, MLX 8-bit quantization
- LoRA on top 16 of 28 transformer layers
- AdamW, LR  $10^{-5}$ , batch size 1, max seq 2048
- 200 iterations, 83 train + 10 valid
- Hardware: Apple M4 Pro, 48 GB unified memory
- Wall time: ≈ 7 minutes
- Peak resident memory: 12.7 GB
- Trainable parameters: 2.6M / 7.6B (0.034 %)

## Eval prompts (held out)

- What is the nature of evil?
- Is faith reasonable?
- What is the relationship between grace and free will?
- How does the soul relate to the body?
- Why does God permit suffering?
- Can a human truly know God?
- What does it mean to love one’s neighbor?
- Is there such a thing as natural law?
- How do you reconcile divine foreknowledge with free will?
- What is the meaning of human life?

None of these appear in the 93 teacher-generated training pairs.

## Results: rubric scores



Four-dimension rubric (0–3 each, total 0–12 per prompt) summed across 10 held-out philosophical questions ( $N = 10$ , max 120). Largest deltas: **scholastic register +17**, **CCC grounding +19**, **structure +10**.

## Qualitative example

Q. How do you reconcile divine foreknowledge with free will?

**BASE (Qwen 2.5 7B-Instruct):** "The reconciliation of divine foreknowledge with free will is a classic philosophical and theological problem that has been debated for centuries. This issue arises from the apparent tension between the idea that a divine being has perfect knowledge of all future events..."

**SFT (Qwen 2.5 7B + LoRA on 83 scholastic pairs):** "This is perhaps the most ancient and most pressing question in all of metaphysics, and it is one that the Angelic Doctor addresses with great precision.

**Objection 1.** It would seem that divine foreknowledge destroys free will, for if God knows..."

## Takeaway

Small open-weights models can be moved into a niche historical register cheaply when capacity is sufficient (7B not 1.5B) and quantization is gentle (Q8 not Q4). The bottleneck is not the optimization recipe but the *teacher signal*: a stronger model generating tightly-prompted training pairs.

## Phase 2: scaling & voice rebalancing

Phase 2 scales to 377 training pairs (~4× Phase 1) with per-source register hints (Aquinas form for *Summa*/CCC chunks, Augustinian for *Confessions*/*City of God*), 800 SFT iterations. Best checkpoint: **iter 400** (mild overfitting after).

**Augustinian gap closed:** **aug** dim 3 → 7–13. The model now switches register per prompt rather than forcing every answer into Summa form.

**Strict rubric undervalues this** (penalizes Augustinian answers on **scholastic\_register**). We add a **register\_fluency** = **max(reg, aug)** metric and a *balanced total*; under it, Phase 2 reaches **68/90 vs Phase 1’s 66/90**.

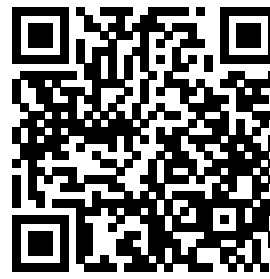
## Negative result: DPO saturation

The Phase 2 DPO chain (chosen = SFT-v2 output, rejected = base output) was inert. From iteration 1: val loss **0.000**, val accuracy **1.000**, chosen/rejected margin **35.4 nats**. The SFT policy already separated chosen from rejected with overwhelming margin, so DPO had no gradient signal.

*Lesson.* Within-model preference data is degenerate: when the policy generated chosen and the base generated rejected, the DPO objective is satisfied at initialization. Useful DPO signal requires preference pairs the policy has not already pre-learned to favor or disfavor.

## Resources

- Repo:** [github.com/pleyva2004/scholastic-llm](https://github.com/pleyva2004/scholastic-llm)
- Paper:** [pleyva2004.github.io/scholastic-llm/main.pdf](https://pleyva2004.github.io/scholastic-llm/main.pdf)
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